

Rigorous Derivation of Emerging Dynamics in Background-Independent Qubit Networks

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1 Introduction to the Global Null Space

The fundamental premise of our model is the existence of a global state $|\Psi\rangle$ in the total Hilbert space $\mathcal{H}_{tot} = \mathcal{H}_C \otimes \mathcal{H}_S$. For this mapping to be mathematically valid, the total Hamiltonian $\hat{H}_{tot}$ must possess a non-trivial kernel (Null Space):

$$\text{Ker}(\hat{H}_{tot}) = \{|\Psi\rangle \in \mathcal{H}_{tot} \mid \hat{H}_{tot} |\Psi\rangle = 0\} \neq \emptyset. \quad (1)$$

Proposition 1.1. *Condition (1) holds if and only if the intersection of the energy spectra satisfies*

$$\text{Spec}(\hat{H}_C) \cap \text{Spec}(-\hat{H}_S) \neq \emptyset. \quad (2)$$

Proof. Let $\{E_k^C, |e_k\rangle_C\}$ and $\{E_j^S, |f_j\rangle_S\}$ be complete eigensystems of $\hat{H}_C$ and $\hat{H}_S$, respectively. Since $\hat{H}_{tot} = \hat{H}_C \otimes I_S + I_C \otimes \hat{H}_S$, the product basis $\{|e_k\rangle_C \otimes |f_j\rangle_S\}_{k,j}$ diagonalises $\hat{H}_{tot}$:

$$\hat{H}_{tot}(|e_k\rangle_C \otimes |f_j\rangle_S) = (E_k^C + E_j^S)(|e_k\rangle_C \otimes |f_j\rangle_S).$$

($\Rightarrow$) If $E_{k_0}^C + E_{j_0}^S = 0$, then $|e_{k_0}\rangle_C \otimes |f_{j_0}\rangle_S \neq 0$ is a non-trivial element of $\text{Ker}(\hat{H}_{tot})$.

($\Leftarrow$) Let $|\Psi\rangle = \sum_{k,j} c_{kj} |e_k\rangle_C \otimes |f_j\rangle_S \neq 0$ satisfy $\hat{H}_{tot} |\Psi\rangle = 0$. Then $\sum_{k,j} c_{kj} (E_k^C + E_j^S) |e_k\rangle_C \otimes |f_j\rangle_S = 0$. By linear independence of the product basis, $c_{kj} (E_k^C + E_j^S) = 0$ for all k, j . Since $|\Psi\rangle \neq 0$ there exists (k_0, j_0) with $c_{k_0 j_0} \neq 0$, forcing $E_{k_0}^C = -E_{j_0}^S \in \text{Spec}(\hat{H}_C) \cap \text{Spec}(-\hat{H}_S)$. $\square$

In an N -qubit simulation, condition (2) is achieved by tuning the local bias terms to ensure $E_C^k + E_S^j = 0$.

2 Clock Representation and Stone's Theorem

We define the *clock basis* $|t\rangle_C$ through the unitary group generated by $\hat{H}_C$. By Stone's theorem on one-parameter unitary groups, the strongly continuous map $t \mapsto e^{-i\hat{H}_C t}$ has $\hat{H}_C$ as its unique self-adjoint generator:

$$|t\rangle_C = U(t) |0\rangle_C = e^{-i\hat{H}_C t} |0\rangle_C. \quad (3)$$

Taking the Hilbert-space adjoint of (3) gives $\langle t|_C = \langle 0|_C e^{i\hat{H}_C t}$. Differentiating with respect to t :

$$\frac{d}{dt} \langle t|_C = \langle 0|_C \cdot i\hat{H}_C e^{i\hat{H}_C t} = \langle 0|_C e^{i\hat{H}_C t} \cdot i\hat{H}_C = \langle t|_C (i\hat{H}_C), \quad (4)$$

where the second equality uses $[e^{i\hat{H}_C t}, \hat{H}_C] = 0$; this commutator vanishes because $e^{i\hat{H}_C t} = \sum_{n=0}^{\infty} (i\hat{H}_C t)^n / n!$ and every power of $\hat{H}_C$ commutes with $\hat{H}_C$. Multiplying (4) by i :

$$i \frac{d}{dt} \langle t|_C = -\langle t|_C \hat{H}_C. \quad (5)$$

3 Step-by-Step Flow of Unitary Emergence

We now derive the internal Schrödinger equation from the static WdW equation.

Step 1: Projection of the Global Constraint

Starting from the WdW equation $\hat{H}_{tot} |\Psi\rangle = 0$, we project onto the clock state $\langle t|_C$:

$$\langle t|_C (\hat{H}_C \otimes I_S + I_C \otimes \hat{H}_S) |\Psi\rangle = 0. \quad (6)$$

Step 2: Separation of Operators

Expanding (6) by linearity:

$$\langle t|_C \hat{H}_C |\Psi\rangle + \langle t|_C (I_C \otimes \hat{H}_S) |\Psi\rangle = 0. \quad (7)$$

To show that $\hat{H}_S$ passes through the clock projection, write $|\Psi\rangle = \sum_\alpha |\phi_\alpha\rangle_C \otimes |\chi_\alpha\rangle_S$. Then

$$\langle t|_C (I_C \otimes \hat{H}_S) |\Psi\rangle = \sum_\alpha \langle t|\phi_\alpha|t|\phi_\alpha\rangle_C \hat{H}_S |\chi_\alpha\rangle_S = \hat{H}_S \left(\sum_\alpha \langle t|\phi_\alpha|t|\phi_\alpha\rangle_C |\chi_\alpha\rangle_S \right) = \hat{H}_S (\langle t|_C |\Psi\rangle),$$

where the second equality uses the $\mathbb{C}$ -linearity of $\hat{H}_S$ on $\mathcal{H}_S$ (the scalar coefficients $\langle t|\phi_\alpha|t|\phi_\alpha\rangle_C \in \mathbb{C}$ factor out). Equation (7) becomes:

$$\langle t|_C \hat{H}_C |\Psi\rangle + \hat{H}_S (\langle t|_C |\Psi\rangle) = 0. \quad (8)$$

Step 3: Identification of the Relative State

Define the *relative (conditional) state* $|\psi(t)\rangle_S \equiv \langle t|_C |\Psi\rangle$. Since $|\Psi\rangle$ is the time-independent WdW state, the time derivative passes through as:

$$i \frac{d}{dt} |\psi(t)\rangle_S = i \frac{d}{dt} (\langle t|_C |\Psi\rangle) = \left(i \frac{d}{dt} \langle t|_C \right) |\Psi\rangle \stackrel{(5)}{=} -\langle t|_C \hat{H}_C |\Psi\rangle. \quad (9)$$

Rearranging (9) gives $\langle t|_C \hat{H}_C |\Psi\rangle = -i \frac{d}{dt} |\psi(t)\rangle_S$. Substituting into (8):

$$-i \frac{d}{dt} |\psi(t)\rangle_S + \hat{H}_S |\psi(t)\rangle_S = 0. \quad (10)$$

Step 4: Final Derivation

Rearranging (10) yields the standard Schrödinger form:

$$i \frac{d}{dt} |\psi(t)\rangle_S = \hat{H}_S |\psi(t)\rangle_S. \quad (11)$$

This is the central result of the Page–Wootters mechanism: unitary time evolution emerges from the timeless WdW constraint $\hat{H}_{tot} |\Psi\rangle = 0$ via clock–system entanglement.

4 Mathematical Consistency Check: The Pauli Limit

In finite-dimensional qubit systems, the spectrum of $\hat{H}_C$ is discrete. Two important consequences follow.

Lemma 4.1 (Time Resolution). *For a clock Hamiltonian $\hat{H}_C$ with operator norm $\|\hat{H}_C\|$, the minimum resolvable time interval satisfies*

$$\Delta t \geq \frac{1}{2\|\hat{H}_C\|}, \quad \text{i.e. } \Delta t \sim 1/\|\hat{H}_C\|.$$

Proof. The Mandelstam–Tamm energy-time uncertainty relation states that for any observable $\hat{A}$:

$$\Delta E_C \cdot \tau_A \geq \frac{1}{2} |\langle [\hat{H}_C, \hat{A}] \rangle|,$$

where $\tau_A = \Delta A / |\langle \dot{\hat{A}} \rangle|$ is the characteristic evolution time of $\hat{A}$. Taking $\hat{A}$ to be the projector onto the initial clock state $|0\rangle_C$, τ_A equals the minimum time for the clock to become distinguishable from its initial value. Since $\Delta E_C \leq \|\hat{H}_C\|$, we obtain $\tau_A \geq (2\|\hat{H}_C\|)^{-1}$, i.e. $\Delta t \equiv \tau_A \sim 1/\|\hat{H}_C\|$. $\square$

Lemma 4.2 (Poincaré Recurrence). *The relative state $|\psi(t)\rangle_S$ satisfying (11) is quasi-periodic with recurrence time $T \leq 2\pi/\Delta E_{\min}$, where $\Delta E_{\min} := \min_{k \neq j} |E_k^S - E_j^S|$.*

Proof. Expand in the energy eigenbasis of $\hat{H}_S$:

$$|\psi(t)\rangle_S = \sum_j a_j e^{-iE_j^S t} |f_j\rangle_S, \quad a_j = \langle f_j | \psi(0) \rangle_S.$$

This is a finite sum of oscillatory terms (spectrum discrete by assumption). The state returns near its initial value when all relative phases satisfy $e^{-i(E_k^S - E_j^S)T} \approx 1$ simultaneously. The slowest cycle is governed by $\Delta E_{\min}$, requiring $\Delta E_{\min} \cdot T = 2\pi$, hence $T \leq 2\pi/\Delta E_{\min}$. $\square$

1. **Discretization:** By Lemma 4.1, time t is sampled at resolutions $\Delta t \sim 1/\|\hat{H}_C\|$.

2. **Poincaré Recurrence:** By Lemma 4.2, the state $|\psi(t)\rangle_S$ returns to its initial value after $T = 2\pi/\Delta E_{\min}$.

This ‘error’ is not a logical flaw but a physical consequence of finite information density (Bekenstein bound) in qubit-based universes.

5 Noise as Graph Perturbation

In the BI-QMN framework, the system is represented by a weighted graph $G = (V, E, W)$, where V is the set of qubits and W_{ij} represents the mutual information (or entanglement measure) between qubits i and j .

Definition 5.1 (Graph Weight Function). Given the N -qubit density matrix ρ , define the edge weight between qubits $i \neq j$ as $W_{ij} = |\rho_{ij}|$ (magnitude of off-diagonal coherence). The degree matrix is $D_{ii} = \sum_{j \neq i} W_{ij}$, and the Graph Laplacian is

$$L = D - W = \sum_{(i,j) \in E} W_{ij} (e_i - e_j)(e_i - e_j)^T, \quad (12)$$

where $e_k \in \mathbb{R}^{|V|}$ is the k -th standard basis vector.

Under a Lindblad decoherence channel, off-diagonal elements decay as $\rho_{ij}(t) = \rho_{ij}(0)e^{-\gamma_{ij}t}$, so for a time step Δt :

$$\delta W_{ij} = W_{ij}(0)(1 - e^{-\gamma_{ij}\Delta t}) \approx W_{ij}(0)\gamma_{ij}\Delta t > 0.$$

Each weight reduction contributes the rank-one perturbation $-\delta W_{ij}(e_i - e_j)(e_i - e_j)^T$ to L , giving

$$L_{\text{noise}} = L_0 + \Delta L, \quad \Delta L = - \sum_{(i,j) \in E} \delta W_{ij} (e_i - e_j)(e_i - e_j)^T. \quad (13)$$

Since each rank-one summand is positive semidefinite and $\delta W_{ij} > 0$, ΔL is *negative semidefinite* ($\Delta L \preceq 0$).

6 Weyl's Inequality and Spectral Shift

Theorem 6.1 (Weyl's Inequality). *Let A and B be $n \times n$ Hermitian matrices with ordered eigenvalues. Then for all $i = 1, \dots, n$:*

$$\lambda_i(A) + \lambda_{\min}(B) \leq \lambda_i(A + B) \leq \lambda_i(A) + \lambda_{\max}(B).$$

Applying Theorem 6.1 with $A = L_0$ and $B = \Delta L$, the perturbed eigenvalues $\tilde{\lambda}_i$ satisfy:

$$\lambda_i(L_0) + \lambda_{\min}(\Delta L) \leq \tilde{\lambda}_i \leq \lambda_i(L_0) + \lambda_{\max}(\Delta L). \quad (14)$$

Because $\Delta L \preceq 0$, we have $\lambda_{\max}(\Delta L) \leq 0$, so the upper bound gives $\tilde{\lambda}_i \leq \lambda_i(L_0)$: *all eigenvalues shift downward*. Furthermore $\text{tr}(\Delta L) = -\sum_{i,j} \delta W_{ij} < 0$ for any non-zero noise, which forces $\lambda_{\min}(\Delta L) < 0$. The magnitude of the shift is bounded below by

$$|\lambda_{\min}(\Delta L)| = \|\Delta L\|_2 \geq \frac{1}{\sqrt{n}} \|\Delta L\|_F > 0,$$

where we used $\|B\|_2 \geq \|B\|_F / \sqrt{n}$ for Hermitian B . This confirms that any non-zero noise ΔL induces a deterministic, detectable shift in the spectral density, providing a mathematical basis for anomaly detection.

7 Error Differentiation: X vs. Z Errors

We hypothesize that different quantum error channels map to distinct types of structural distortions.

7.1 Bit-Flip (X) Errors as Edge Deletion

An X -error on qubit k reduces the effective entanglement with its neighbors, decreasing weights $\{W_{kj}\}_{j \sim k}$ by $\{\delta W_{kj}\}$.

Proposition 7.1 (First-Order Spectral Shift under X -Errors). *Under an X -error that reduces edge weights $\{W_{kj}\}$ by $\{\delta W_{kj} > 0\}$, the first-order shift of the m -th Laplacian eigenvalue is*

$$\delta \lambda_m = - \sum_{j \sim k} \delta W_{kj} ((v_m)_k - (v_m)_j)^2 \leq 0, \quad (15)$$

where $\mathbf{v}_m$ is the m -th normalised eigenvector of L_0 (assuming non-degeneracy).

Proof. From Eq. (12), edge (k, j) contributes $W_{kj}(e_k - e_j)(e_k - e_j)^T$ to L . A weight reduction by δW_{kj} induces the elementary perturbation $\Delta L_{(k,j)} = -\delta W_{kj}(e_k - e_j)(e_k - e_j)^T$. By first-order Rayleigh–Schrödinger perturbation theory for non-degenerate eigenvalues:

$$\begin{aligned} \delta \lambda_m &= \mathbf{v}_m^T \Delta L \mathbf{v}_m = - \sum_{j \sim k} \delta W_{kj} \mathbf{v}_m^T (e_k - e_j)(e_k - e_j)^T \mathbf{v}_m \\ &= - \sum_{j \sim k} \delta W_{kj} [(v_m)_k - (v_m)_j]^2. \end{aligned}$$

Non-negativity of the squared terms and positivity of δW_{kj} ensure $\delta \lambda_m \leq 0$. □

For high-frequency eigenvectors $\mathbf{v}_m$ (large m), rapid spatial oscillation ensures $|(v_m)_k - (v_m)_j| = \mathcal{O}(1)$, so the aggregate shift across all error edges is:

$$\Delta \lambda_{\text{large}} \propto - \sum_{(i,j) \in \text{error}} \delta W_{ij}. \quad (16)$$

7.2 Phase-Flip (Z) Errors as Topological Warp

Z -errors disrupt the phase relationship $\langle t|_C|\Psi\rangle$, causing global connectivity loss. This is captured by the *Fiedler value* λ_2 , which bounds the edge expansion of G via the Cheeger inequality:

$$\frac{\lambda_2}{2} \leq h(G) \leq \sqrt{2\lambda_2 \lambda_{\max}(L)},$$

where $h(G)$ is the Cheeger constant. The perturbed Fiedler value is given by the Courant–Fischer min-max theorem:

$$\tilde{\lambda}_2 = \min_{\substack{x \perp \mathbf{1} \\ \|x\|=1}} \frac{x^T (L_0 + \Delta L) x}{x^T x}, \quad (17)$$

where $\mathbf{1} \in \mathbb{R}^n$ denotes the all-ones vector. Since $\Delta L \preceq 0$, we have $x^T \Delta L x \leq 0$ for all x , therefore

$$\tilde{\lambda}_2 \leq \min_{\substack{x \perp \mathbf{1} \\ \|x\|=1}} x^T L_0 x = \lambda_2.$$

The gap $\lambda_2 - \tilde{\lambda}_2 \geq 0$ directly quantifies the loss of algebraic connectivity (global phase coherence). A significant drop indicates *Phase Collapse*.

8 Relational Spectral Detection Without External Time

8.1 Clock-Conditioned Relative States

We do not assume an external background time parameter. Instead, following the Page–Wootters construction, we consider a timeless global state $|\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S$ satisfying the global constraint

$$\hat{H}_{\text{tot}}|\Psi\rangle = 0. \quad (18)$$

Let $\{|\tau\rangle_C\}$ denote the clock basis. The corresponding clock-conditioned relative state is defined as

$$|\psi(\tau)\rangle_S := {}_C\langle\tau|\Psi\rangle, \quad (19)$$

or, in density-matrix form,

$$\rho_S(\tau) := \frac{\text{Tr}_C[(|\tau\rangle\langle\tau|_C \otimes I_S) |\Psi\rangle\langle\Psi|]}{\text{Tr}[(|\tau\rangle\langle\tau|_C \otimes I_S) |\Psi\rangle\langle\Psi|]}. \quad (20)$$

Here, τ is not an external time variable, but an internal relational label induced by the clock subsystem.

8.2 Relational Graph and Laplacian Family

For each clock-conditioned state $\rho_S(\tau)$, we construct a weighted graph

$$G_\tau = (V, E, W^{(\tau)}), \quad (21)$$

where the edge weights are determined from the relational structure of $\rho_S(\tau)$.

For example, one may define

$$W_{ij}^{(\tau)} := f(\rho_S(\tau); i, j), \quad (22)$$

where f denotes a pairwise relational measure such as coherence magnitude, mutual information, or another correlation functional.

The corresponding degree matrix is

$$D_{ii}^{(\tau)} = \sum_{j \neq i} W_{ij}^{(\tau)}, \quad (23)$$

and the clock-conditioned graph Laplacian is

$$L(\tau) = D^{(\tau)} - W^{(\tau)}. \quad (24)$$

Thus, instead of a time-evolving Laplacian, we obtain a relationally indexed family of Laplacians

$$\{L(\tau)\}_{\tau \in \mathcal{T}}, \quad (25)$$

where $\mathcal{T}$ denotes the clock-label domain.

8.3 Relational Spectral Density

Let

$$\Lambda(\tau) = \{\lambda_1(\tau), \dots, \lambda_n(\tau)\} \quad (26)$$

be the eigenvalue set of $L(\tau)$.

We define the clock-conditioned empirical spectral density (ESD) as

$$p_\tau(\lambda) = \frac{1}{n} \sum_{k=1}^n \delta(\lambda - \lambda_k(\tau)), \quad (27)$$

or, numerically, via a kernel density estimator,

$$p_\tau(\lambda) = \frac{1}{n} \sum_{k=1}^n \frac{1}{\sqrt{2\pi\sigma_h^2}} \exp\left(-\frac{(\lambda - \lambda_k(\tau))^2}{2\sigma_h^2}\right). \quad (28)$$

Accordingly, spectral variation is interpreted as variation across clock-conditioned relational slices, rather than evolution over an external time axis.

8.4 Reference and Observed Relational Spectra

For each relational slice τ , let $p_\tau^{\text{ref}}(\lambda)$ denote the reference spectral density and $p_\tau^{\text{obs}}(\lambda)$ the observed spectral density under perturbation.

The slice-wise information-theoretic discrepancy is defined as

$$D_{\text{KL}}^{(\tau)} := D_{\text{KL}}\left(p_\tau^{\text{ref}} \parallel p_\tau^{\text{obs}}\right). \quad (29)$$

Theorem 8.1 (Relational Spectral Detectability). *For a fixed clock-conditioned slice τ , if the perturbation induces a non-trivial spectral deformation, $\|\Delta L(\tau)\|_F > 0$, then $D_{\text{KL}}^{(\tau)} > 0$.*

Proof. If $\|\Delta L(\tau)\|_F > 0$, then by Weyl's inequality (Theorem 6.1) at least one eigenvalue of $L(\tau)$ is shifted relative to the reference Laplacian. Hence $p_\tau^{\text{ref}}(\lambda) \neq p_\tau^{\text{obs}}(\lambda)$ on a set of non-zero measure. By Gibbs' inequality, $D_{\text{KL}}(p_\tau^{\text{ref}} \parallel p_\tau^{\text{obs}}) > 0$. $\square$

Thus, $D_{\text{KL}}^{(\tau)}$ serves as a sufficient certificate of non-trivial spectral deformation within the relational slice indexed by τ .

8.5 Relationally Averaged Detection Functional

To compare the entire relational family rather than a single slice, we define the aggregated detection functional

$$\mathcal{D}_{\text{KL}} = \int_{\mathcal{T}} D_{\text{KL}}\left(p_\tau^{\text{ref}} \parallel p_\tau^{\text{obs}}\right) d\mu(\tau), \quad (30)$$

where $\mu(\tau)$ is a measure over the clock-label domain $\mathcal{T}$.

In discrete simulations, this reduces to

$$\mathcal{D}_{\text{KL}} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} D_{\text{KL}}^{(\tau)}. \quad (31)$$

This quantity measures the global mismatch between two relational spectral families without introducing an external time parameter.

8.6 Hybrid Detection Metric with Geometric Augmentation

While $D_{\text{KL}}^{(\tau)}$ provides an information-theoretic detection certificate, it does not directly quantify the geometric magnitude of spectral displacement. To incorporate geometric sensitivity, we define

$$D_{\text{detect}}^{(\tau)} = \alpha \tilde{D}_{\text{KL}}^{(\tau)} + (1 - \alpha) \widetilde{W}_1^{(\tau)}, \quad (32)$$

where

$$\tilde{D}_{\text{KL}}^{(\tau)} = \text{Norm} \left[D_{\text{KL}} \left(p_{\tau}^{\text{ref}} \parallel p_{\tau}^{\text{obs}} \right) \right] \quad (33)$$

and

$$\widetilde{W}_1^{(\tau)} = \text{Norm} \left[W_1 \left(p_{\tau}^{\text{ref}}, p_{\tau}^{\text{obs}} \right) \right]. \quad (34)$$

Here, W_1 denotes the Wasserstein-1 distance between the two clock-conditioned spectral densities, and $\alpha \in [0, 1]$ controls the balance between information-theoretic sensitivity and geometric displacement.

We then define the relationally aggregated hybrid score

$$\mathcal{D}_{\text{detect}} = \int_{\mathcal{T}} D_{\text{detect}}^{(\tau)} d\mu(\tau), \quad (35)$$

or, in discrete form,

$$\mathcal{D}_{\text{detect}} = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} D_{\text{detect}}^{(\tau)}. \quad (36)$$

Remark 8.2. The KL term preserves the information-theoretic detectability guarantee at each relational slice, while the Wasserstein term quantifies geometric spectral displacement across the clock-conditioned family.

8.7 Symmetric Stability Check

For numerical robustness, we optionally introduce the Jensen–Shannon divergence at each relational slice:

$$D_{\text{JS}}^{(\tau)} = \frac{1}{2} D_{\text{KL}} \left(p_{\tau}^{\text{ref}} \parallel m_{\tau} \right) + \frac{1}{2} D_{\text{KL}} \left(p_{\tau}^{\text{obs}} \parallel m_{\tau} \right), \quad (37)$$

with

$$m_{\tau} = \frac{1}{2} \left(p_{\tau}^{\text{ref}} + p_{\tau}^{\text{obs}} \right). \quad (38)$$

The relationally averaged Jensen–Shannon score is

$$\mathcal{D}_{\text{JS}} = \int_{\mathcal{T}} D_{\text{JS}}^{(\tau)} d\mu(\tau), \quad (39)$$

or its discrete counterpart. This quantity is used as a symmetric numerical stability check, rather than as the primary detectability certificate.

8.8 Localized Geometric Error Differentiation

To distinguish different structural perturbation types, we introduce a band-limited Wasserstein descriptor on a spectral region $\mathcal{B}$:

$$W_1^{(\tau, \mathcal{B})} = \int_{\mathcal{B}} \left| F_{\tau}^{\text{ref}}(\lambda) - F_{\tau}^{\text{obs}}(\lambda) \right| d\lambda, \quad (40)$$

where F_{τ}^{ref} and F_{τ}^{obs} are the corresponding cumulative spectral distributions.

This enables local-versus-global deformation analysis within each relational slice:

- high-frequency bands capture localized structural perturbations,
- low-frequency bands capture global connectivity loss.

Thus, error discrimination is formulated as a geometric comparison across relational spectral slices rather than as a trajectory over background time.

8.9 Detection Criterion

An anomaly is declared whenever $\mathcal{D}_{\text{detect}} > \varepsilon$, where ε is determined from the background distribution of relationally aggregated divergence values under unperturbed fluctuations:

$$\varepsilon = \mathbb{E}[\mathcal{D}_{\text{vac}}] + \kappa \text{Std}[\mathcal{D}_{\text{vac}}]. \quad (41)$$

This criterion remains fully relational and does not require any external time indexing.

9 Relational Spectral Trajectory Manifold

9.1 Clock-Conditioned Spectral Family

Let $|\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S$ be a timeless global state satisfying

$$\hat{H}_{\text{tot}}|\Psi\rangle = 0. \quad (42)$$

For each clock label $\tau \in \mathcal{T}$, define the clock-conditioned relative state

$$\rho_S(\tau) := \frac{\text{Tr}_C[(|\tau\rangle\langle\tau|_C \otimes I_S) |\Psi\rangle\langle\Psi|]}{\text{Tr}[(|\tau\rangle\langle\tau|_C \otimes I_S) |\Psi\rangle\langle\Psi|]}. \quad (43)$$

From each relational slice $\rho_S(\tau)$, construct the weighted graph Laplacian

$$L(\tau) = D^{(\tau)} - W^{(\tau)}, \quad (44)$$

and let its ordered eigenvalue vector be

$$\Lambda(\tau) = (\lambda_1(\tau), \lambda_2(\tau), \dots, \lambda_n(\tau)) \in \mathbb{R}^n. \quad (45)$$

We define the *relational spectral family* associated with $|\Psi\rangle$ as

$$\Gamma_\Psi : \mathcal{T} \rightarrow \mathbb{R}^n, \quad \Gamma_\Psi(\tau) := \Lambda(\tau). \quad (46)$$

Thus, Γ_Ψ is not a trajectory over external time, but an ordered family of spectral signatures indexed by the internal clock labels.

9.2 Spectral Trajectory Space

Let $\mathcal{M}_{\text{spec}}$ denote the set of admissible relational spectral families,

$$\mathcal{M}_{\text{spec}} := \{\Gamma : \mathcal{T} \rightarrow \mathbb{R}^n \mid \Gamma(\tau) = \Lambda(L(\rho_S(\tau))) \text{ for some physically admissible } \rho_S(\tau)\}. \quad (47)$$

We interpret $\mathcal{M}_{\text{spec}}$ as the *relational spectral trajectory manifold*, i.e., the manifold of all physically realizable clock-conditioned spectral families.

Remark 9.1. Each element of $\mathcal{M}_{\text{spec}}$ is a full relational trajectory Γ , not a single spectral snapshot.

Remark 9.2. Strictly speaking, $\mathcal{M}_{\text{spec}}$ is treated here as a metric trajectory space of admissible spectral families rather than as a fully established smooth manifold. The geodesic formulation in Secs. 9.3–9.9 should therefore be interpreted as a formal geometric model. The discrete representation in Eq. (84) provides the computational realization, in which $\mathcal{M}_{\text{spec}}$ reduces to a constrained subset of $\mathbb{R}^{mn}$.

9.3 Tangent Structure

Given a smooth one-parameter family of relational trajectories

$$\Gamma_s \in \mathcal{M}_{\text{spec}}, \quad s \in (-\epsilon, \epsilon), \quad (48)$$

with $\Gamma_0 = \Gamma$, the tangent vector at Γ is defined by

$$V(\tau) = \left. \frac{d}{ds} \Gamma_s(\tau) \right|_{s=0} \in \mathbb{R}^n. \quad (49)$$

Hence, the tangent space at Γ is

$$T_\Gamma \mathcal{M}_{\text{spec}} = \left\{ V : \mathcal{T} \rightarrow \mathbb{R}^n \mid V(\tau) = \left. \frac{d}{ds} \Gamma_s(\tau) \right|_{s=0} \text{ for some smooth family } \Gamma_s \right\}. \quad (50)$$

Intuitively, V describes an infinitesimal deformation of the full relational spectral family.

9.4 Riemannian Metric on the Trajectory Manifold

To quantify distances between relational trajectories, we introduce a Riemannian metric on $\mathcal{M}_{\text{spec}}$.

For tangent vectors $U, V \in T_\Gamma \mathcal{M}_{\text{spec}}$, define

$$g_\Gamma(U, V) = \int_{\mathcal{T}} U(\tau)^\top G(\Gamma(\tau)) V(\tau) d\mu(\tau), \quad (51)$$

where $G(\Gamma(\tau))$ is a positive-definite local metric tensor on $\mathbb{R}^n$, and μ is a measure over the clock-label domain $\mathcal{T}$.

A simple Euclidean choice is

$$G(\Gamma(\tau)) = I_n, \quad (52)$$

which yields

$$g_\Gamma(U, V) = \int_{\mathcal{T}} U(\tau)^\top V(\tau) d\mu(\tau). \quad (53)$$

More generally, one may choose G to weight low-frequency and high-frequency spectral components differently.

9.5 Trajectory Length and Distance

Let $\gamma : [0, 1] \rightarrow \mathcal{M}_{\text{spec}}$ be a smooth path in the trajectory manifold, where each $\gamma(s)$ is itself a relational spectral family:

$$\gamma(s) = \Gamma_s. \quad (54)$$

Its velocity field is

$$\dot{\gamma}(s) = \frac{d}{ds} \Gamma_s \in T_{\Gamma_s} \mathcal{M}_{\text{spec}}. \quad (55)$$

The length of γ is defined by

$$\text{Len}(\gamma) = \int_0^1 \sqrt{g_{\Gamma_s}(\dot{\gamma}(s), \dot{\gamma}(s))} ds. \quad (56)$$

The induced geodesic distance between two relational trajectories $\Gamma^{(a)}, \Gamma^{(b)} \in \mathcal{M}_{\text{spec}}$ is

$$d_{\mathcal{M}}(\Gamma^{(a)}, \Gamma^{(b)}) = \inf_{\gamma(0)=\Gamma^{(a)}, \gamma(1)=\Gamma^{(b)}} \text{Len}(\gamma). \quad (57)$$

9.6 Observed and Reference Trajectories

Let

$$\Gamma_{\text{ref}} \in \mathcal{M}_{\text{spec}} \quad (58)$$

denote the reference relational spectral trajectory, and let

$$\Gamma_{\text{obs}} \in \mathcal{M}_{\text{spec}} \quad (59)$$

denote the observed trajectory under perturbation.

The trajectory-level distortion is then defined as

$$\Delta_{\text{traj}} = d_{\mathcal{M}}(\Gamma_{\text{ref}}, \Gamma_{\text{obs}}). \quad (60)$$

This quantity measures the global deformation of the entire clock-conditioned spectral family, rather than the discrepancy at any single relational slice.

9.7 Geodesic Recovery Principle

We now formulate recovery as a geometric restoration problem on $\mathcal{M}_{\text{spec}}$.

Given an observed trajectory Γ_{obs} , define the recovered trajectory as

$$\Gamma^* = \arg \min_{\Gamma \in \mathcal{A}} d_{\mathcal{M}}(\Gamma, \Gamma_{\text{obs}}), \quad (61)$$

where $\mathcal{A} \subset \mathcal{M}_{\text{spec}}$ is the admissible set of physically consistent or reference-compatible trajectories.

Equivalently, if Γ_{ref} is known, the recovery path is the minimizing geodesic

$$\gamma^* = \arg \min_{\gamma(0)=\Gamma_{\text{obs}}, \gamma(1)=\Gamma_{\text{ref}}} \text{Len}(\gamma). \quad (62)$$

This gives the central interpretation:

Recovery is not modeled as evolution over background time, but as the shortest admissible path on the relational spectral trajectory manifold.

9.8 Geodesic Equation

Let the path be locally represented by coordinates

$$x^a(s), \quad a = 1, \dots, \dim(\mathcal{M}_{\text{spec}}), \quad (63)$$

with Riemannian metric $g_{ab}(x)$. The geodesic equation is

$$\frac{d^2 x^a}{ds^2} + \Gamma_{bc}^a(x) \frac{dx^b}{ds} \frac{dx^c}{ds} = 0, \quad (64)$$

where the Christoffel symbols are

$$\Gamma_{bc}^a = \frac{1}{2} g^{ad} (\partial_b g_{dc} + \partial_c g_{db} - \partial_d g_{bc}). \quad (65)$$

In the Euclidean metric case $G = I_n$, the connection is flat and the geodesic reduces to a straight interpolation in trajectory space:

$$\Gamma_s(\tau) = (1-s)\Gamma_{\text{obs}}(\tau) + s\Gamma_{\text{ref}}(\tau), \quad s \in [0, 1]. \quad (66)$$

9.9 Variational Formulation

Equivalently, geodesics are stationary points of the action

$$\mathcal{S}[\gamma] = \frac{1}{2} \int_0^1 g_{\Gamma_s}(\dot{\gamma}(s), \dot{\gamma}(s)) ds. \quad (67)$$

Thus, the geodesic recovery problem can be formulated as

$$\gamma^* = \arg \min_{\gamma} \mathcal{S}[\gamma], \quad (68)$$

subject to the endpoint conditions

$$\gamma(0) = \Gamma_{\text{obs}}, \quad \gamma(1) = \Gamma_{\text{ref}}, \quad (69)$$

or more generally,

$$\gamma(1) \in \mathcal{A}. \quad (70)$$

10 Admissible Relational Spectral Trajectory Set

10.1 Definition of the Admissible Set

Let $\mathcal{M}_{\text{spec}}$ denote the relational spectral trajectory manifold. We define the admissible set $\mathcal{A} \subset \mathcal{M}_{\text{spec}}$ as

$$\mathcal{A} = \mathcal{A}_{\text{lap}} \cap \mathcal{A}_{\text{rel}} \cap \mathcal{A}_{\text{ref}}, \quad (71)$$

where each component enforces a distinct class of physical and structural constraints.

10.2 Laplacian Consistency

A trajectory $\Gamma \in \mathcal{M}_{\text{spec}}$ belongs to $\mathcal{A}_{\text{lap}}$ if for all $\tau \in \mathcal{T}$, the associated matrix $L(\tau)$ satisfies

$$L(\tau) = L(\tau)^\top, \quad (72)$$

$$L(\tau) \succeq 0, \quad (73)$$

$$L(\tau)\mathbf{1} = 0, \quad (74)$$

$$L_{ij}(\tau) \leq 0 \quad (i \neq j), \quad (75)$$

$$L_{ii}(\tau) = \sum_{j \neq i} -L_{ij}(\tau). \quad (76)$$

Thus,

$$\mathcal{A}_{\text{lap}} = \{\Gamma \in \mathcal{M}_{\text{spec}} \mid L(\tau) \text{ is a valid graph Laplacian for all } \tau\}. \quad (77)$$

10.3 Relational Consistency

A trajectory Γ is relationally admissible if it satisfies

(i) Slice Regularity

$$\int_{\mathcal{T}} \left\| \frac{d}{d\tau} \Gamma(\tau) \right\|^2 d\mu(\tau) < \infty. \quad (78)$$

(ii) Spectral Ordering

$$\lambda_1(\tau) \leq \lambda_2(\tau) \leq \dots \leq \lambda_n(\tau) \quad \forall \tau \in \mathcal{T}. \quad (79)$$

(iii) **Clock Consistency** Let $d_{\text{clock}}(\rho_S(\tau))$ be a clock-consistency measure. Then

$$\int_{\mathcal{T}} d_{\text{clock}}(\rho_S(\tau)) d\mu(\tau) \leq \eta. \quad (80)$$

Thus,

$$\mathcal{A}_{\text{rel}} = \{\Gamma \in \mathcal{A}_{\text{lap}} \mid \Gamma \text{ is slice-regular and clock-consistent}\}. \quad (81)$$

10.4 Reference Compatibility

Let $\mathfrak{R} \subset \mathcal{M}_{\text{spec}}$ be a reference trajectory family. We define

$$\mathcal{A}_{\text{ref}} = \left\{ \Gamma \in \mathcal{A}_{\text{rel}} \mid \inf_{\Gamma' \in \mathfrak{R}} d_{\mathcal{M}}(\Gamma, \Gamma') \leq \delta \right\}. \quad (82)$$

10.5 Final Definition

The admissible set is given by

$$\mathcal{A} = \left\{ \Gamma \in \mathcal{M}_{\text{spec}} \mid \begin{array}{l} L(\tau) \text{ is a valid Laplacian for all } \tau, \\ \Gamma \text{ is slice-regular and clock-consistent,} \\ \inf_{\Gamma' \in \mathfrak{R}} d_{\mathcal{M}}(\Gamma, \Gamma') \leq \delta \end{array} \right\}. \quad (83)$$

10.6 Discrete Computational Form

Let $\mathcal{T} = \{\tau_1, \dots, \tau_m\}$. Then a trajectory is represented as

$$\mathbf{z}_{\Gamma} = \begin{bmatrix} \Lambda(\tau_1) \\ \Lambda(\tau_2) \\ \vdots \\ \Lambda(\tau_m) \end{bmatrix} \in \mathbb{R}^{mn}. \quad (84)$$

The trajectory distance becomes

$$d_{\mathcal{M}}(\Gamma^{(a)}, \Gamma^{(b)}) = \left(\sum_{r=1}^m w_r \left\| \Lambda^{(a)}(\tau_r) - \Lambda^{(b)}(\tau_r) \right\|_2^2 \right)^{1/2}. \quad (85)$$

10.7 Penalty-Based Admissibility

Define

$$\Phi_{\text{lap}}(\Gamma) = \sum_{r=1}^m \left(\|L(\tau_r) - L(\tau_r)^{\top}\|_F^2 + \|\min(0, \lambda(L(\tau_r)))\|_2^2 + \|L(\tau_r)\mathbf{1}\|_2^2 \right), \quad (86)$$

$$\Phi_{\text{smooth}}(\Gamma) = \sum_{r=1}^{m-1} \frac{\|\Lambda(\tau_{r+1}) - \Lambda(\tau_r)\|_2^2}{(\tau_{r+1} - \tau_r)^2}, \quad (87)$$

$$\Phi_{\text{ref}}(\Gamma) = \min_{\Gamma' \in \mathfrak{R}} d_{\mathcal{M}}(\Gamma, \Gamma')^2, \quad (88)$$

$$\Phi_{\text{clock}}^{\text{geom}}(\Gamma) = \sum_{r=1}^{m-1} \frac{\|\Lambda(\tau_{r+1}) - \Lambda(\tau_r)\|_2}{d_{\text{clock}}^{\text{geom}}(\tau_r, \tau_{r+1}) + \epsilon} \quad (89)$$

Here Φ_{lap} is a graph-validity surrogate, Φ_{smooth} is a regularizer on relational variation, and Φ_{clock} is a relational consistency penalty. The quantity Φ_{ref} remains useful as a coarse reference-compatibility diagnostic, but it does not by itself provide a directed restoration force toward the clean family.

We therefore define the relaxed admissible set

$$\mathcal{A}_\epsilon = \{\Gamma : \Phi_{\text{lap}} \leq \epsilon_1, \Phi_{\text{smooth}} \leq \epsilon_2, \Phi_{\text{ref}} \leq \epsilon_3, \Phi_{\text{clock}} \leq \epsilon_4\}. \quad (90)$$

10.8 Recovery Objective

The earlier recovery objective primarily enforced admissibility and observational consistency, but did not sufficiently encode restoration toward the reference family. To address this mismatch, we introduce an explicit reference-guided recovery term based on the nearest admissible reference anchor. This modification separates restoration from feasibility, allowing the optimization to favor trajectories that are both admissible and structurally closer to the clean reference family.

Let

$$\Gamma_{\text{ref,anchor}} := \arg \min_{\Gamma' \in \mathcal{R}} d_{\mathcal{M}}(\Gamma', \Gamma_{\text{obs}})^2 \quad (91)$$

denote the nearest reference anchor to the observed trajectory.

The recovered trajectory is then defined by the restoration-aware objective

$$\Gamma^* = \arg \min_{\Gamma} \left[\alpha d_{\mathcal{M}}(\Gamma, \Gamma_{\text{obs}})^2 + \beta d_{\mathcal{M}}(\Gamma, \Gamma_{\text{ref,anchor}})^2 + \lambda_1 \Phi_{\text{lap}}(\Gamma) + \lambda_2 \Phi_{\text{smooth}}(\Gamma) + \lambda_3 \Phi_{\text{clock}}(\Gamma) \right]. \quad (92)$$

In this formulation, the observed-fit term preserves consistency with the measured trajectory, the anchor term supplies an explicit restoration direction toward the reference family, and the remaining penalties serve only as feasibility / regularization controls.

10.9 Two-Stage Admissible Recovery

We refine the recovery procedure by introducing a two-stage structure on the admissible trajectory space.

Stage 1 (Discrete Admissible Projection). Given an observed trajectory Γ_{obs} , we first project onto a discrete admissible candidate bank $\mathcal{A}_{\text{bank}} \subset \mathcal{A}$:

$$\Gamma^{(1)} = \arg \min_{\Gamma \in \mathcal{A}_{\text{bank}}} \left[\alpha d_M(\Gamma, \Gamma_{\text{obs}})^2 + \beta d_M(\Gamma, \Gamma_{\text{ref,anchor}})^2 + \lambda_1 \Phi_{\text{lap}}(\Gamma) + \lambda_2 \Phi_{\text{smooth}}(\Gamma) + \lambda_3 \Phi_{\text{clock}}(\Gamma) \right].$$

This step provides a stable admissible anchor while explicitly preserving restoration toward the nearest compatible reference family.

Stage 2 (Admissible Convex Refinement). We optionally perform a continuous refinement within a local convex hull of admissible candidates. Let $\{\Gamma_k\}$ denote a selected subset of admissible candidates. We consider convex combinations

$$\Gamma(\alpha) = \sum_k \alpha_k \Gamma_k, \quad \alpha_k \geq 0, \quad \sum_k \alpha_k = 1,$$

restricted to those α such that $\Gamma(\alpha) \in \mathcal{A}$.

The refined trajectory is defined as

$$\Gamma^{(2)} = \arg \min_{\Gamma(\alpha) \in \mathcal{A}} \left[\alpha d_M(\Gamma(\alpha), \Gamma_{\text{obs}})^2 + \beta d_M(\Gamma(\alpha), \Gamma_{\text{ref,anchor}})^2 + \lambda_1 \Phi_{\text{lap}}(\Gamma(\alpha)) + \lambda_2 \Phi_{\text{smooth}}(\Gamma(\alpha)) + \lambda_3 \Phi_{\text{clock}}(\Gamma(\alpha)) \right].$$

Interpretation. Stage 1 performs a coarse projection onto an admissible discrete set, while Stage 2 carries out a local refinement under the same restoration-aware objective. The reference-anchor term is the actual recovery driver, whereas Φ_{lap} , Φ_{smooth} , and Φ_{clock} retain their roles as feasibility / regularization terms.

Practical Use. Stage 1 is used as the default recovery due to its stability and interpretability. Stage 2 is treated as an optional diagnostic refinement and is adopted only when it decreases the restoration-aware objective while maintaining admissibility. In practice this means checking

$$J_{\text{rec}}(\Gamma^{(2)}) < J_{\text{rec}}(\Gamma^{(1)})$$

together with the required admissibility constraints, e.g.

$$\Phi_{\text{lap}}(\Gamma^{(2)}) \leq \epsilon_1, \quad \Phi_{\text{clock}}(\Gamma^{(2)}) \leq \epsilon_4.$$

Default Operating Point. For the paper-facing default configuration, the discrete reference bank is taken to have width

$$\Delta_{\text{bank}} = 20^\circ$$

around the central reference phase, and the restoration weight is set to

$$\beta = 16.$$

This operating point is chosen as the smallest anchor strength that already gives a stable Stage 1 surrogate in the validation sweep. Narrower banks such as 5° are treated as unstable narrow-bank regimes, while very large anchor weights such as $\beta = 128$ are treated as high-anchor regimes that overly suppress useful Stage 2 diagnostic variation. Wider operating points or stronger anchors are therefore relegated to sensitivity / ablation tables rather than used as the main paper default.

10.10 Hybrid Policy Layer for Encoded Baselines

The encoded-QEC experiments produce two recovery candidates for each sampled noise instance:

$$\begin{aligned} A &:= \text{syndrome-first decoded recovery,} \\ B &:= \text{relational recovery from the admissible restoration pipeline.} \end{aligned}$$

Candidate A represents the conventional decoder-facing baseline. Candidate B represents the relational restoration mechanism developed above: it is selected from the admissible trajectory family using the restoration-aware objective rather than from the syndrome prior alone. The hybrid layer is therefore not a third recovery algorithm. It is a decision layer that asks when the relational candidate is reliable enough to replace the syndrome candidate.

Candidate diagnostics

For each candidate $X \in \{A, B\}$ we record:

$$F_X, \quad \Delta F_X := F_X - F_{\text{before}}, \quad J_X := J_{\text{rec}}(\Gamma_X),$$

together with the admissibility report produced by the relational trajectory constraints. Let $\mathcal{K}$ be the set of active admissibility penalties, and let θ_k denote the corresponding threshold for penalty Φ_k . The normalized admissibility violation used by the robust gate is

$$v_X := \max_{k \in \mathcal{K}} \max \left\{ 0, \frac{\Phi_k(\Gamma_X) - \theta_k}{\theta_k + \epsilon} \right\}.$$

Thus $v_X = 0$ means that all active constraints are within threshold, while positive values measure the worst normalized excursion beyond the admissible boundary. This is strictly more informative than a Boolean admissible/inadmissible flag, and it matches the implementation used by the robust gate.

Decision variables used by the hybrid layer

The policies below are deterministic maps from the same row-level diagnostics to one of the two candidates. Let

$$a_X := \mathbf{1}[X \text{ is admissible}], \quad d_X := d(\Gamma_X, \Gamma_{\text{obs}}), \quad u_X := \frac{1}{1 + J_X}.$$

The syndrome support score used in the current implementation is

$$s_A := p_{\text{obs}}(0), \quad s_B := 0,$$

where $p_{\text{obs}}(0)$ is the observed mean no-error syndrome probability. This encodes the prior that the decoder-facing candidate A is directly supported by the syndrome record, while the relational candidate B must earn selection through trajectory proximity, restoration objective, and admissibility. Under the imperfect syndrome protocols, the degraded syndrome variables enter this policy layer; the physical construction of A and B is otherwise kept fixed so that the experiment isolates decision risk rather than changing the recovery candidates themselves.

C1: rule-based conservative hybrid

The first policy is the original conservative hybrid used as a reference point. Let c_{obs} denote observed syndrome consistency and let τ_J be the objective tie tolerance. In the seed-10 experiments

$$\tau_J = 10^{-9}, \quad c_{\text{obs}} \text{ is required for the objective tie switch.}$$

The implemented decision rule is

$$C_1 = B \iff [(a_A = 0) \wedge (a_B = 1)] \vee [(a_B = 1) \wedge (J_B < J_A - \tau_J) \wedge c_{\text{obs}}].$$

Otherwise $C_1 = A$. Thus C_1 is not a fitted score model. It has two explicit routes to B : veto a non-admissible syndrome candidate when B is admissible, or use B as an objective tie-breaker only when the observed syndrome still appears consistent.

C2: balanced score proposal policy

The second policy is the main score-based proposal rule. For weights $\lambda^{(2)} = (\lambda_s, \lambda_t, \lambda_i, \lambda_o)$ define

$$S_X^{(2)} = \lambda_s s_X - \lambda_t d_X - \lambda_i (1 - a_X) + \lambda_o u_X.$$

The configuration used in the current seed-10 experiments is

$$\lambda^{(2)} = (1.0, 1.0, 2.0, 0.5).$$

The rule is

$$C_2 = \begin{cases} B, & S_B^{(2)} > S_A^{(2)}, \\ A, & S_B^{(2)} < S_A^{(2)}. \end{cases}$$

When admissibility differs and the two scores are exactly tied, the implementation breaks the tie toward the admissible candidate; otherwise exact ties preserve A . This small tie convention is included only for determinism. The substantive point is that C_2 is the policy that may actively trade syndrome support against relational objective quality, trajectory proximity, and admissibility penalty.

C3: hard-admissibility ablation

The original C_3 should now be interpreted as an ablation of the hypothesis that a hard admissibility filter defines a distinct safety-first operating zone. Its rule is

$$C_3 = \begin{cases} A, & a_A = 1, a_B = 0, \\ B, & a_A = 0, a_B = 1, \\ \arg \max_{X \in \{A, B\}} S_X^{(3)}, & a_A = a_B, \end{cases}$$

with ties in the last line resolved toward A . In the equal-admissibility case the implementation uses

$$S_X^{(3)} = 0.8 s_X - 1.2 d_X + 0.3 u_X.$$

The configured inadmissibility weight for the C_3 score template is 5.0, but the hard admissibility branch removes the penalty term once both candidates have the same admissibility status. This is precisely why C_3 is not the final redesign: it tests a Boolean version of safety, whereas the observed failure mode is a downside-risk question inside imperfect syndrome evidence.

C3R: robust veto layer on C2

The final policy used by the current code is C_3^R , not C_3 . It treats C_2 as a proposal policy and applies a one-sided veto to proposed switches to B . Let

$$\Delta S := S_B^{(2)} - S_A^{(2)}, \quad \Delta v := v_A - v_B.$$

Imperfect syndrome information is summarized by

$$u_{\text{raw}} = (1 - r_{\text{obs}}) + p_{\text{noise}} + a_{\text{amb}} + p_{\text{meas}} + p_{\text{reset}} + \ell_{\text{syn}},$$

$$u_{\text{syn}} = \text{clip}_{[0,1]}(u_{\text{raw}}),$$

where r_{obs} is the syndrome observation ratio, p_{noise} is the syndrome-bit corruption probability, a_{amb} is the ambiguity level, p_{meas} and p_{reset} are measurement/reset error probabilities, and ℓ_{syn} is measured syndrome information loss. The raw value is kept as a diagnostic; the clipped value is used by the gate.

The robust gate is

$$C_3^R = B \iff [C_2 = B] \wedge [\Delta S \geq \delta_S] \wedge [v_A \geq \delta_A] \wedge [a_B = 1, v_B \leq \delta_B] \wedge [u_{\text{syn}} \leq u_{\text{max}}].$$

Otherwise $C_3^R = A$. The current configuration is

$$\delta_S = 0, \quad \delta_A = 0, \quad \delta_B = 0, \quad u_{\text{max}} = 0.99.$$

With these defaults the v_A condition is non-restrictive because $v_A \geq 0$ by definition; it is retained as a tuning handle for stricter future gates. The two active safety requirements are that B itself be admissible and that the clipped syndrome uncertainty not be saturated. Thus $u_{\text{syn}} = 1$ is treated as too degraded to trust a relational switch.

Equivalently, define

$$G_R := \mathbf{1}[C_2 = B] \mathbf{1}[\Delta S \geq \delta_S] \mathbf{1}[v_A \geq \delta_A] \mathbf{1}[a_B = 1, v_B \leq \delta_B] \mathbf{1}[u_{\text{syn}} \leq u_{\text{max}}].$$

Then

$$C_3^R = \begin{cases} B, & G_R = 1, \\ A, & G_R = 0. \end{cases}$$

This gives the immediate invariants

$$C_3^R = B \implies C_2 = B, \quad \chi_B(C_3^R) \leq \chi_B(C_2),$$

where $\chi_B(C)$ is the indicator that policy C selects B . The robust layer can only block C_2 switches; it never creates additional switches that C_2 rejected.

Why C_3^R replaces C_3 . The negative clean-map result for C_3 is not the main evidence for the redesign. It is only an ablation result showing that Boolean admissibility alone does not create a useful new policy region once inadmissibility has already been priced into C_2 . The actual redesign is the move from a hard admissibility selector to a one-sided downside-risk veto on C_2 . This matters because the relevant failure mode is not “choose the admissible candidate”; it is “avoid trusting a relational switch when the available syndrome evidence is too degraded to justify the downside risk.” A structurally admissible B can still lower logical fidelity, and an observed syndrome that looks consistent can hide a true failure boundary under partial or corrupted observation. C_3^R is therefore the paper-facing policy, while C_3 is kept as a diagnostic ablation.

10.11 Evaluation Metrics for Robust Switching

The clean-syndrome regime map is useful for identifying whether a policy has an independent operating zone. It is not sufficient for judging C_3^R , because C_3^R is primarily an imperfect-information safety layer. We therefore separate policy quality, downside-risk, and switch-intervention metrics.

Policy quality metrics

For each policy $C \in \{C_1, C_2, C_3, C_3^R\}$:

$$\overline{\Delta F}_C, \quad P(\Delta F_C \geq 0), \quad P(F_C \geq F_{\text{success}}), \quad P(C = B)$$

measure mean fidelity gain, non-worsening rate, logical-success rate, and relational candidate selection rate. These are performance metrics, not by themselves robust safety metrics: a policy may improve the mean while worsening the lower tail.

False-safe and tail-risk metrics

The structural false-safe rate measures whether a policy chooses a structurally unsafe candidate while the observed syndrome appears consistent:

$$\text{FS}_{\text{struct}}(C) = P[\text{observed syndrome consistent} \wedge C \text{ inadmissible}].$$

This metric is useful, but it misses a failure mode that matters directly for recovery: a switch can be structurally admissible but still lower logical fidelity. We therefore also use the fidelity false-safe rate

$$\text{FS}_F(C) = P[\Delta F_C < -\varepsilon_F],$$

with ε_F equal to the experiment’s fidelity margin.

For magnitude-sensitive downside risk, define the empirical loss excess

$$R_-(C) := \mathbb{E}[\max\{0, -\Delta F_C - \varepsilon_F\}].$$

We also track lower-tail statistics of the gain distribution,

$$q_{.05}(C) := \inf\{x : \hat{P}(\Delta F_C \leq x) \geq 0.05\},$$

$$\text{CVaR}_{.05}(C) := \hat{\mathbb{E}}[\Delta F_C \mid \Delta F_C \leq q_{.05}(C)].$$

The intended conservative claim is therefore not that C_3^R uniformly dominates C_2 . The claim is that the gate is explicitly parameterized to reduce the frequency and magnitude of downside switches when syndrome information becomes unreliable, and that this reduction should be visible in FS_F , $q_{.05}$, $\text{CVaR}_{.05}$, and the intervention diagnostics below.

C3-R switch-intervention metrics

Since C_3^R is only active when C_2 proposes B , the primitive block indicator is

$$B_R := \mathbf{1}[C_2 = B] \mathbf{1}[C_3^R = A].$$

Two candidate-level ground-truth comparisons are then recorded:

$$H_2 := \mathbf{1}[C_2 = B] \mathbf{1}[\Delta F_A > \Delta F_B + \varepsilon_F],$$

$$G_2 := \mathbf{1}[C_2 = B] \mathbf{1}[\Delta F_B > \Delta F_A + \varepsilon_F].$$

Here H_2 means that the proposed C_2 switch to B was harmful relative to staying with A , while G_2 means that the proposed switch was beneficial. The central conditional rates are

$$P(B_R = 1 \mid C_2 = B), \quad P(H_2 = 1 \mid B_R = 1), \quad P(B_R = 1 \mid H_2 = 1),$$

and

$$P(B_R = 1 \mid G_2 = 1), \quad P(B_R = 0 \mid G_2 = 1).$$

We call these the block rate among C_2 switches, harmful-block precision, harmful-switch recall, beneficial-switch block rate, and beneficial-switch retention. This makes explicit that C_3^R is not a universal harmful-switch oracle: it is an uncertainty-triggered safety gate that trades off prevented losses against missed beneficial switches.

The corresponding magnitude decomposition is

$$L_{\text{prev}} := \sum_i B_R^{(i)} \max\{0, \Delta F_A^{(i)} - \Delta F_B^{(i)}\},$$

$$G_{\text{miss}} := \sum_i B_R^{(i)} \max\{0, \Delta F_B^{(i)} - \Delta F_A^{(i)}\},$$

$$G_{\text{net}} := L_{\text{prev}} - G_{\text{miss}} = \sum_i \left(\Delta F_{C_3^R}^{(i)} - \Delta F_{C_2}^{(i)} \right).$$

Thus $G_{\text{net}} > 0$ says only that, on the sampled grid, blocked harmful-switch losses outweighed missed beneficial-switch gains. It is a sample-level intervention diagnostic, not a theorem of dominance.

Oracle regret and failure-boundary metrics

For each row define the two-candidate oracle gain

$$\Delta F^* := \max\{\Delta F_A, \Delta F_B\}.$$

The oracle regret of policy C is

$$\text{Reg}(C) := \Delta F^* - \Delta F_C.$$

Mean, 95th-percentile, and upper-tail regret summarize how far the policy falls from the best available candidate on the same row. This is a stricter diagnostic than comparing only C_2 and C_3^R .

Finally, we distinguish observed and true failure boundaries:

$$\text{FB}_{\text{obs}}(C) := P[\text{observed syndrome consistent} \wedge \neg \text{logical success}_C],$$

$$\text{FB}_{\text{true}}(C) := P[\text{true syndrome consistent} \wedge \neg \text{logical success}_C].$$

The observed boundary is what the policy can directly see. The true boundary is an offline diagnostic used to check whether observed consistency is hiding a logical failure under partial, noisy, or ambiguous syndrome information.

10.12 Experimental Stress Protocols

The hybrid layer is evaluated under five increasingly degraded syndrome regimes.

- **Clean regime map.** Complete, uncorrupted syndrome observations. This tests whether a policy has an independent operating zone when syndrome information is reliable.
- **Partial syndrome.** Only a fraction r_{obs} of syndrome information is exposed to the policy layer. This stresses uncertainty from missing syndrome information.
- **Noisy syndrome.** Syndrome bits are corrupted with probability p_{noise} . This stresses random observational corruption while keeping the observation ratio fixed.
- **Partial+noisy syndrome.** Missing and corrupted syndrome information are combined. This is the most direct stress test for over-trusting relational switches under degraded evidence.
- **Ambiguity+measurement/reset syndrome.** Syndrome ambiguity and physical measurement/reset error probabilities are swept jointly. This isolates whether the robust gate reacts to ambiguity-driven uncertainty rather than only to missing bits.

The clean regime map and the imperfect-syndrome maps answer different questions. The clean map asks whether a policy defines a distinct operating zone. The imperfect maps ask whether a policy avoids downside risk when the evidence for switching has become less trustworthy.

10.13 Empirical Operating Zones of C_1 , C_2 , C_3 , and C_3^R

In the clean-syndrome seed-10 regime map, the full encoded grid is aggregated over 240 regime cells after averaging over seeds. The preferred-policy counts are

$$\#(C_1) = 197, \quad \#(C_2) = 43, \quad \#(C_3) = 0, \quad \#(C_3^R) = 0.$$

Thus C_1 remains the dominant default policy, C_2 occupies a nontrivial local operating zone, and neither C_3 nor C_3^R forms an independent region on the clean-syndrome grid. This is expected for C_3^R : when syndrome observation is complete and uncorrupted, the raw uncertainty is zero, the uncertainty gate is open, and the robust policy collapses to the same choices as C_2 .

The C_2 -preferred cells form a structured cluster rather than a scattered set of exceptions. All 43 C_2 -preferred clean-regime cells occur under the phaseflip code, and they are concentrated in phase-sensitive stochastic sectors:

$$16 \text{ dephasing}, \quad 10 \text{ phaseflip}, \quad 9 \text{ mixed Pauli}, \quad 6 \text{ depolarizing}, \quad 2 \text{ bitflip}.$$

In these C_2 -preferred cells the mean logical-fidelity gains satisfy

$$\overline{\Delta F}_{C_1} \approx 0.0135, \quad \overline{\Delta F}_{C_2} \approx 0.0250,$$

while the mean rates of selecting the relational candidate are

$$\overline{\chi_B(C_1)} \approx 0.9349, \quad \overline{\chi_B(C_2)} \approx 0.4000.$$

The fidelity false-safe rates in the same cells are

$$\overline{\text{FS}_F(C_1)} \approx 0.2674, \quad \overline{\text{FS}_F(C_2)} \approx 0.1093.$$

Hence the improvement of C_2 comes from switching more selectively, not from always trusting the relational candidate. The rule-based C_1 policy tends to over-select B near phase-sensitive boundaries where the syndrome baseline should still be preserved.

10.14 Final Seed-10 C3-R Results

The final seed-10 rerun was evaluated under the clipped uncertainty convention and $u_{\max} = 0.99$. The resulting policy-level outcomes are:

Regime	n	$\overline{\Delta F}_{C_2}$	$\overline{\Delta F}_{C_3^R}$	$P_{\text{succ}}(C_2)$	$P_{\text{succ}}(C_3^R)$	$\text{FS}_F(C_2)$	$\text{FS}_F(C_3^R)$	$q_{.05}(C_2)$	$q_{.05}(C_3^R)$
clean	2400	0.1273	0.1273	0.6358	0.6358	0.0238	0.0238	0.0000	0.0000
partial	2880	0.1169	0.1189	0.4552	0.5976	0.0694	0.0389	-0.0152	-0.0006
noisy	3600	0.1169	0.1169	0.4556	0.4556	0.0678	0.0678	-0.0104	-0.0104
partial+noisy	2160	0.1170	0.1212	0.4551	0.7241	0.0685	0.0134	-0.0104	0.0000
ambiguity+meas.	3600	0.1600	0.1604	0.5208	0.5978	0.0000	0.0000	0.0192	0.0238

The lower-tail and oracle diagnostics give the same qualitative picture:

Regime	$\text{CVaR}_{.05}(C_2)$	$\text{CVaR}_{.05}(C_3^R)$	$\overline{\text{Reg}}(C_2)$	$\overline{\text{Reg}}(C_3^R)$	$\text{FB}_{\text{true}}(C_2)$	$\text{FB}_{\text{true}}(C_3^R)$
clean	-0.0033	-0.0033	0.0132	0.0132	0.2375	0.2375
partial	-0.0208	-0.0147	0.0271	0.0251	0.3611	0.2653
noisy	-0.0207	-0.0207	0.0270	0.0270	0.3589	0.3589
partial+noisy	-0.0206	-0.0034	0.0271	0.0228	0.3616	0.1819
ambiguity+meas.	0.0078	0.0084	0.0336	0.0332	0.2136	0.1847

The switch-intervention table is the most direct evidence for the redesigned policy:

Regime	$C_2 = B$	blocked	$P(B_R C_2 = B)$	precision	recall	ben. block	ben. retain	G_{net}
clean	1529	0	0.0000	—	—	—	—	0.0000
partial	1094	476	0.4351	0.8697	0.4461	0.3750	0.6250	5.6616
noisy	1370	0	0.0000	—	—	—	—	0.0000
partial+noisy	821	670	0.8161	0.8791	0.8475	0.6739	0.3261	9.1212
ambiguity+meas.	895	357	0.3989	0.7759	0.3929	0.4211	0.5789	1.5749

Interpretation. The clean map confirms that C_3^R has no separate clean-syndrome operating zone; it reduces to C_2 because the uncertainty evidence does not ask for a veto. The noisy-only map also leaves C_3^R inactive. This is not a failure of the implementation: under this stress protocol the raw uncertainty mean is only 0.038, far below the saturation threshold, so the robust gate has no reason to block C_2 switches.

The partial and partial+noisy maps are the main positive evidence. In the partial map, C_3^R blocks 43.51% of C_2 switches, with harmful-block precision 86.97% and net intervention gain 5.6616. In the partial+noisy map, the effect is stronger: C_3^R blocks 81.61% of C_2 switches, recovers the lower tail from $q_{.05} = -0.0104$ to 0.0000, improves $\text{CVaR}_{.05}$ from -0.0206 to -0.0034 , and reduces the true failure boundary from 0.3616 to 0.1819.

The ambiguity+measurement map shows why the old false-safe-only evaluation was too narrow. Both C_2 and C_3^R have $\text{FS}_F = 0$, but the robust gate still slightly improves mean gain, logical success, lower-tail gain, oracle regret, and true failure-boundary rate. These improvements are modest rather than decisive, so this regime should be described as supporting evidence for the safety-gate interpretation, not as a separate dominance claim.

The main cost is also visible. C_3^R sometimes blocks beneficial C_2 switches: the beneficial-switch block rate is 37.50% in the partial map, 67.39% in the partial+noisy map, and 42.11% in the ambiguity+measurement map. Thus the conservative paper-facing statement should be that C_3^R is a downside-risk policy whose sampled net intervention gain is positive in the degraded syndrome regimes tested here. It should not yet be stated as an optimal or uniformly dominant switching policy.

Taxonomy after the redesign. The present interpretation is:

- C_1 : default rule-based hybrid on most clean regimes,
- C_2 : balanced phase-sensitive correction that suppresses over-switching,
- C_3 : hard-admissibility probe with no independent observed operating zone,

- C_3^R : no independent clean-syndrome zone, but a dedicated imperfect-information safety layer that vetoes selected C_2 switches under degraded syndrome evidence.

The role of C_3^R is therefore not to outperform C_2 in every clean or low-uncertainty setting. Its role is to reduce downside-risk exposure when the evidence for switching becomes partial, ambiguous, or jointly corrupted.

11 Clock Geometry and Relational Consistency

11.1 Clock Orbit and Local Distinguishability

Let the clock states be generated by

$$|\tau\rangle_C = e^{-i\hat{H}_C\tau}|0\rangle_C, \quad (93)$$

where $\hat{H}_C$ is the clock Hamiltonian.

We define the local distinguishability between adjacent clock labels as

$$d_{\text{clock}}^{\text{geom}}(\tau, \tau + \delta\tau) := 1 - |\langle \tau | \tau + \delta\tau \rangle_C|^2. \quad (94)$$

Equivalently, using the Bures distance,

$$D_B(\tau, \tau + \delta\tau) = \sqrt{2(1 - \sqrt{F_\tau})}, \quad (95)$$

where

$$F_\tau = F(|\tau\rangle_C \langle \tau|_C, |\tau + \delta\tau\rangle_C \langle \tau + \delta\tau|_C) = |\langle \tau | \tau + \delta\tau \rangle_C|^2. \quad (96)$$

For sufficiently small $\delta\tau$, we obtain the expansion

$$d_{\text{clock}}^{\text{geom}}(\tau, \tau + \delta\tau) = (\Delta H_C)_\tau^2 (\delta\tau)^2 + O((\delta\tau)^4), \quad (97)$$

where

$$(\Delta H_C)_\tau^2 = \langle \tau | \hat{H}_C^2 | \tau \rangle - \langle \tau | \hat{H}_C | \tau \rangle^2. \quad (98)$$

Interpretation The quantity $d_{\text{clock}}^{\text{geom}}$ measures the intrinsic distinguishability of adjacent clock labels and induces a local geometric structure on the clock orbit. It does not depend on external time and serves as an intrinsic resolution measure of the relational clock.

11.2 Relational Response Between Clock Slices

Let $\rho_S(\tau)$ be the clock-conditioned system state. We define the relational response between adjacent slices as

$$d_{\text{resp}}(\tau, \tau + \delta\tau) := d(\rho_S(\tau), \rho_S(\tau + \delta\tau)), \quad (99)$$

In the main formulation, we instantiate d_{resp} as the spectral response

$$d_{\text{resp}}(\tau, \tau + \delta\tau) := \|\Lambda(\tau + \delta\tau) - \Lambda(\tau)\|_2,$$

thereby aligning the clock-consistency penalty with the geometry of the spectral trajectory space. The fidelity-based alternative is discussed in the appendix as a robustness check.

11.3 Clock Consistency Measure

We define the clock consistency ratio as

$$\mathcal{C}_{\text{clock}}(\tau) = \frac{d_{\text{resp}}(\tau, \tau + \delta\tau)}{d_{\text{clock}}^{\text{geom}}(\tau, \tau + \delta\tau) + \epsilon}, \quad (100)$$

where $\epsilon > 0$ is a small regularization constant.

Interpretation This ratio measures whether the variation of the system state is consistent with the intrinsic resolution of the clock.

- $\mathcal{C}_{\text{clock}}(\tau) \ll 1$: stable relational encoding
- $\mathcal{C}_{\text{clock}}(\tau) \approx 1$: consistent evolution
- $\mathcal{C}_{\text{clock}}(\tau) \gg 1$: anomalous or inconsistent slicing

11.4 Trajectory-Level Clock Consistency

We extend the definition to the full trajectory:

$$\Phi_{\text{clock}}(\Gamma) = \int_{\mathcal{T}} \mathcal{C}_{\text{clock}}(\tau) d\mu(\tau), \quad (101)$$

or in discrete form,

$$\Phi_{\text{clock}}(\Gamma) = \sum_{r=1}^{m-1} \frac{d(\rho_S(\tau_r), \rho_S(\tau_{r+1}))}{d_{\text{clock}}^{\text{geom}}(\tau_r, \tau_{r+1}) + \epsilon}. \quad (102)$$

A trajectory is considered clock-consistent if

$$\Phi_{\text{clock}}(\Gamma) \leq \eta. \quad (103)$$

Remark 11.1. The admissibility threshold η is calibrated statistically from the background distribution of Φ_{clock} evaluated on unperturbed reference trajectories, whereas the slice spacing $\delta\tau$ is constrained by the clock resolution bound of Lemma 4.1. These two quantities serve distinct roles and are not directly related.

The abstract clock penalty Φ_{clock} introduced in Eq. (89) is concretely instantiated in Sec. 11 through the clock-geometry-based consistency ratio in Eqs. (99)–(101).